

Analytical and Numerical Solutions to the Black-Scholes Equation

Conceptual Foundations, Exact Derivations, and Monte Carlo Methods

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Agenda

- 1 Stochastic Calculus Foundations
- 2 Deriving the PDE
- 3 Analytical Solution
- 4 Numerical Validation

1. Modeling Market Dynamics

The Conceptual Problem:

- Classical mechanics assumes continuous variables are smooth (differentiable).
- Financial asset prices are infinitely jagged due to continuous information flow.
- Standard calculus ($\frac{dS}{dt}$) fails. We must separate movement into a deterministic trend and an unpredictable shock.

Geometric Brownian Motion (GBM):

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \implies dS_t = \mu S_t dt + \sigma S_t dW_t$$

- μ : Deterministic drift (expected return).
- σdW_t : Stochastic diffusion driven by a Standard Wiener Process.
- Formatting as percentage return ensures asset prices never cross below zero.

2. Ito's Lemma: Quadratic Variation

Standard Taylor Series Expansion:

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2 + \dots$$

Expanding the $(dS)^2$ term:

$$\begin{aligned} (dS)^2 &= (\mu S dt + \sigma S dW_t)^2 \\ &= \mu^2 S^2 (dt)^2 + 2\mu\sigma S^2 (dt)(dW_t) + \sigma^2 S^2 (dW_t)^2 \end{aligned}$$

As $dt \rightarrow 0$: $(dt)^2 \rightarrow 0$, $(dt)(dW_t) \rightarrow 0$, $(dW_t)^2 \rightarrow dt$

$$\implies (dS)^2 = \sigma^2 S^2 dt$$

Ito's Lemma:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t$$

3. Constructing the Hedged Portfolio

We can construct a locally risk-free portfolio by holding the option (V) and shorting Δ units of the underlying asset (S):

$$\Pi = V - \Delta S \implies d\Pi = dV - \Delta dS$$

Substitute Ito's Lemma and GBM into the portfolio change:

$$d\Pi = (\dots) dt + \sigma S \left(\frac{\partial V}{\partial S} - \Delta \right) dW_t$$

Eliminating Market Risk: By continuously choosing our hedge ratio $\Delta = \frac{\partial V}{\partial S}$, the stochastic dW_t term perfectly cancels out. The portfolio becomes deterministic.

4. The Black-Scholes Partial Differential Equation

After setting $\Delta = \frac{\partial V}{\partial S}$, the portfolio's change is purely deterministic:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

The No-Arbitrage Argument:

Because the hedged portfolio (Π) is strictly free of market risk, it must grow at the exact same rate as a bank account earning the risk-free interest rate (r).

$$d\Pi = r\Pi dt = r \left(V - \frac{\partial V}{\partial S} S \right) dt$$

Equating these two deterministic growth rates yields the foundational PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

5. Analytical Solution: Domain Transformation

The Goal: Eliminate the shifting variable parameters (S, S^2) to map the equation to fundamental physics.

Step 1: Chain Rule Expansion

- Translate to logarithmic moneyness: $x = \ln(S/K)$ (where $K =$ Strike Price)
- Scale time by variance: $\tau = \frac{\sigma^2}{2}(T - t)$

This flattens the unpredictable market onto an infinite number line, yielding an intermediate convection-diffusion equation:

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + (\mathbf{k} - 1) \frac{\partial v}{\partial x} - \mathbf{k}v$$

where $\mathbf{k} = \frac{2r}{\sigma^2}$ (a dimensionless constant combining interest rate and variance).

6. Eliminating Advection and Decay

We apply an integrating factor ($e^{\alpha x + \beta \tau}$) to neutralize the market's natural drift and interest rate decay:

$$v(x, \tau) = e^{\alpha x + \beta \tau} u(x, \tau) \implies \mathbf{u}(\mathbf{x}, \tau) = \mathbf{e}^{-\alpha \mathbf{x} - \beta \tau} \mathbf{v}(\mathbf{x}, \tau)$$

Substituting this into our convection-diffusion equation and grouping the derivatives of u yields:

$$u_\tau = u_{xx} + \underbrace{[2\alpha + (k - 1)]}_{\text{Set to 0}} u_x + \underbrace{[\alpha^2 + \alpha(k - 1) - k - \beta]}_{\text{Set to 0}} u$$

To reduce this to the fundamental Heat Equation ($u_\tau = u_{xx}$), the coefficients of the advection term (u_x) and the decay term (u) must strictly equal zero.

7. Solving for α and β

Solving for α (eliminating advection):

$$2\alpha + k - 1 = 0 \implies \alpha = -\frac{1}{2}(\mathbf{k} - \mathbf{1})$$

Solving for β (eliminating decay):

$$\begin{aligned}\beta &= \alpha^2 + \alpha(k - 1) - k \\ &= \left[-\frac{1}{2}(k - 1)\right]^2 + \left[-\frac{1}{2}(k - 1)\right](k - 1) - k \\ &= -\frac{1}{4}(k - 1)^2 - k \implies \beta = -\frac{1}{4}(\mathbf{k} + \mathbf{1})^2\end{aligned}$$

By selecting these specific constants, the financial parameters vanish, leaving the **1D Heat Equation**:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}$$

8. The Cauchy Problem & Initial Conditions

To solve the Heat Equation ($u_\tau = u_{xx}$), we must establish an initial boundary condition at time $\tau = 0$.

The Transformed Call Payoff (Expression for u):

A European Call option has a payoff of $\max(S - K, 0)$. By applying our domain transformations ($x = \ln(S/K)$ and $u = ve^{-\alpha x}$), the explicit initial condition for u becomes:

$$u(x, 0) = Ke^{-\alpha x} \max(e^x - 1, 0) \equiv f(x)$$

The Point Source Strategy:

Instead of solving this complex piecewise function directly, we first find the fundamental solution for a single point of certainty using the Dirac delta distribution, $\delta(x)$:

$$\text{Fundamental Cauchy Problem: } \begin{cases} u_\tau = u_{xx} \\ u(x, 0) = \delta(x) \end{cases}$$

9. The Fourier Transform

Solving spatial double-derivatives is complex. We use a spatial Fourier Transform to shift the equation into the frequency domain (ω):

$$\mathcal{F}\{u(x, \tau)\} = \hat{u}(\omega, \tau) = \int_{-\infty}^{\infty} u(x, \tau) e^{-i\omega x} dx$$

Converting Calculus to Algebra:

The transform brilliantly converts the spatial double-derivative ($\frac{\partial^2}{\partial x^2}$) into simple algebraic multiplication by $(i\omega)^2$:

$$\frac{\partial \hat{u}}{\partial \tau} = -\omega^2 \hat{u}$$

This basic first-order ODE solves to a simple exponential decay:

$$\hat{u}(\omega, \tau) = e^{-\omega^2 \tau}$$

10. Inverse Transform & Convolution

The Gaussian Heat Kernel (Φ):

The Inverse Fourier Transform of our frequency solution resolves perfectly into a normal probability distribution:

$$\Phi(x, \tau) = \frac{1}{\sqrt{4\pi\tau}} e^{-\frac{x^2}{4\tau}}$$

The Convolution Integral (Superposition):

We treat the Call option payoff ($f(s)$) as an infinite sum of point sources originating at dummy locations s . Since diffusion depends purely on spatial distance, we use the shifted kernel $\Phi(x - s, \tau)$.

$$u(x, \tau) = \int_{-\infty}^{\infty} f(s) \Phi(x - s, \tau) ds$$

Substituting our exact Heat Kernel yields the full expression for u :

$$u(x, \tau) = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} f(s) e^{-\frac{(x-s)^2}{4\tau}} ds$$

11. The Final Analytical Solution

Evaluating the Convolution Integral algebraically yields the exact, closed-form pricing formula for a European Call Option:

The Black-Scholes Formula:

$$C(S, t) = S_t \mathcal{N}(d_1) - Ke^{-r(T-t)} \mathcal{N}(d_2)$$

Where:

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$
$$d_2 = d_1 - \sigma\sqrt{T-t}$$

- $\mathcal{N}(\cdot)$ represents the cumulative distribution function (CDF) of the Standard Normal distribution.

12. Risk-Neutral Dynamics: Applying Ito's Lemma

To numerically simulate the exact path, we assume assets grow at the risk-free rate:

$$dS_t = rS_t dt + \sigma S_t dW_t$$

We cannot integrate this directly. We apply Ito's Lemma to the natural logarithm of the stock price: $f(S_t) = \ln(S_t)$, where $f_S = 1/S_t$ and $f_{SS} = -1/S_t^2$.

$$\begin{aligned} d(\ln S_t) &= \left(\frac{1}{S_t} \right) dS_t - \frac{1}{2} \left(\frac{1}{S_t^2} \right) (dS_t)^2 \\ &= (rdt + \sigma dW_t) - \frac{1}{2} \sigma^2 dt \end{aligned}$$

Grouping terms reveals the true dynamics of the logarithmic return, explicitly showing the "volatility drag" ($-\frac{1}{2}\sigma^2$):

$$d(\ln S_t) = \left(r - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t$$

13. The Exact Simulated Path Equation

We now integrate the logarithmic return from $t = 0$ to expiration (T):

$$\int_0^T d(\ln S_t) = \int_0^T \left(r - \frac{1}{2}\sigma^2 \right) dt + \int_0^T \sigma dW_t$$

Evaluating the integrals yields:

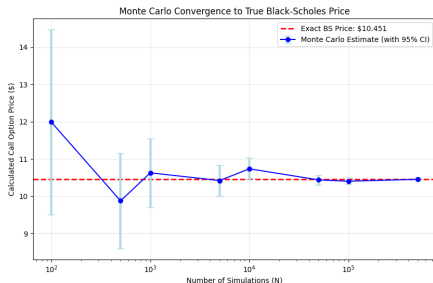
$$\ln(S_T) - \ln(S_0) = \left(r - \frac{1}{2}\sigma^2 \right) T + \sigma W_T$$

By definition, $W_T \sim \mathcal{N}(0, T)$. We can express this as $\sqrt{T}Z$, where Z is a Standard Normal random variable. Exponentiating both sides provides the exact mathematical solution for the final stock price:

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma \sqrt{T}Z \right)$$

14. Python Implementation & Convergence

- Using Python and NumPy vectorization, we simulated 100,000 independent market paths using our derived S_T equation.
- **Result:** As $N \rightarrow 10^5$, the standard error shrinks proportionally to $1/\sqrt{N}$.
- The simulation aggressively converges exactly to the analytical Heat Equation derivation.



15. Conclusion & References

Conclusion:

- Geometric Brownian Motion successfully models non-differentiable asset paths.
- Continuous hedging eliminates stochastic risk, yielding the BS PDE.
- Domain transformations and Fourier Transforms rigorously prove the solution manifests as a Gaussian convolution.
- Computational simulations perfectly validate the analytical mathematical architecture.

Key References:

- 1 Black, F., & Scholes, M. (1973). *The Pricing of Options and Corporate Liabilities*.
- 2 Hull, J. C. (2017). *Options, Futures, and Other Derivatives*.
- 3 Shreve, S. E. (2004). *Stochastic Calculus for Finance II*.